

PAM131440Hz8DCValenceRun1

February 25, 2025

1 Standard Maxx Analysis Template

1.1 Notes

Overview: See below. Note that this run repeated an earlier one but 8% duty cycle instead of 32% (at 40Hz).

Rationale: PAM13.14 driver activation increased number of events without affecting during so we tested for general valence.

Date of Experiment: 2/19/25

Genotypes: PAM13.14>Chrimson and PAM02>wCS

Sex: Females

Light Paradigm: 10 off; 60min on; 10 off; 40Hz, 8% DC, 18V

Design: Two genotypes alternating in groups of three, beginning with Chrimson on the left and Control on the right

Misc: This was the first run of the day, and it started around 10:15am

1.2 Configuration

Configuration was standard. No new or untested code.

```
[1]: #!/autoreload 1
import warnings
from Arena import *
import Tracker
import TwoChoiceTracker
from ExperimentalDesign import *
from Parameters import *

# Suppress all warnings
warnings.filterwarnings('ignore')
```

1.3 Parameters

Just a standard arena max, accepting the default values and a Two-Choice Tracker.

```
[2]: p=Parameters()
p.set_arena_max_values(TrackingType.TWOCHOICETRACKER)
p.print()
```

```

tracking_type: TrackingType.TWOCHOICETRACKER
fps: 0
mm_per_pixel: 0.145
speed_window_seconds: 1
micromove_speed_mm_sec: [0.2, 2]
walking_speed_mm_sec: 2
sleep_threshold_min: 5
interaction_distances: [8]

```

1.4 Read in the data

Next we will read in the data to an Arena object. The format is `Arena(ExperimentName, parameters, data_path="./")`. This will automatically calculate all relevant metrics for the tracking type in question.

The name of the experiment and predefined parameters object are required, but `data_path` is optional (if no data directory is specified, it defaults to the current working directory).

The following files must be present in the defined data directory : * The DTrack experiment file, which is an excel sheet with the name '*ExperimentName.xlsx*'. * The DTrack data files themselves, which are named '*ExperimentName_Data_1.csv*', '*ExperimentName_Data_2.csv*', etc. * An experiment file (most of the time, definitely for a TwoChoiceTracker), which is named '*ExperimentName_Design.txt*'. Note that you can incorporate flipped axes to orient lights for data visualization.

```

[3]: arena = Arena.Arena(p,data_path='./Data/')
      arena.print_short_data_quality_report()

```

	Tracker	HighQuality	NotFound	Indiscernible	StartMinutes	EndMinutes
0	T_0_0	0.978697	0.021303	0.0	0.008481	80.015087
1	T_1_0	0.996388	0.003612	0.0	0.008481	80.015087
2	T_2_0	0.999774	0.000226	0.0	0.008481	80.015087
3	T_3_0	0.999859	0.000141	0.0	0.008481	80.015087
4	T_4_0	0.999859	0.000141	0.0	0.008481	80.015087
5	T_5_0	0.999802	0.000198	0.0	0.008481	80.015087
6	T_6_0	0.999549	0.000451	0.0	0.008481	80.015087
7	T_7_0	0.998815	0.001185	0.0	0.008481	80.015087
8	T_8_0	0.999661	0.000339	0.0	0.008481	80.015087
9	T_9_0	0.999859	0.000141	0.0	0.008481	80.015087
10	T_10_0	0.999379	0.000621	0.0	0.008481	80.015087
11	T_11_0	0.999464	0.000536	0.0	0.008481	80.015087
12	T_12_0	0.998392	0.001608	0.0	0.008481	80.015087
13	T_13_0	0.996727	0.003273	0.0	0.008481	80.015087
14	T_14_0	0.997856	0.002144	0.0	0.008481	80.015087
15	T_15_0	0.999746	0.000254	0.0	0.008481	80.015087
16	T_16_0	0.997404	0.002596	0.0	0.008481	80.015087
17	T_17_0	0.987049	0.012951	0.0	0.008481	80.015087
18	T_18_0	0.970571	0.029429	0.0	0.008481	80.015087
19	T_19_0	0.966536	0.033464	0.0	0.008481	80.015087

20	T_20_0	0.960723	0.039277	0.0	0.008481	80.015087
21	T_21_0	0.972800	0.027200	0.0	0.008481	80.015087
22	T_22_0	0.979092	0.020908	0.0	0.008481	80.015087
23	T_23_0	0.983635	0.016365	0.0	0.008481	80.015087
24	T_24_0	0.993877	0.006123	0.0	0.008481	80.015087
25	T_25_0	0.999605	0.000395	0.0	0.008481	80.015087
26	T_26_0	0.999831	0.000169	0.0	0.008481	80.015087
27	T_27_0	0.999859	0.000141	0.0	0.008481	80.015087
28	T_28_0	0.999295	0.000705	0.0	0.008481	80.015087
29	T_29_0	0.994131	0.005869	0.0	0.008481	80.015087
30	T_30_0	0.983296	0.016704	0.0	0.008481	80.015087
31	T_31_0	0.974013	0.025987	0.0	0.008481	80.015087
32	T_32_0	0.994949	0.005051	0.0	0.008481	80.015087
33	T_33_0	0.978020	0.021980	0.0	0.008481	80.015087
34	T_34_0	0.967636	0.032364	0.0	0.008481	80.015087
35	T_35_0	0.978979	0.021021	0.0	0.008481	80.015087

All data quality is above 0.9.

1.5 Summary

Get the summary results for all trackers, including treatments, etc.

This will copy the results to the clipboard and save the data in a file called '*ExperimentName_Summary.csv*'.

The semicolon at the end of the command will suppress the output to this notebook. Remove it if you want to see the results dataframe.

```
[4]: arena.summarize(copy_to_clipboard=True,write_to_csvfile=True);
```

Or you can also create a faceted summary if you want to separate the different phases of the experiment.

The file name will be '*ExperimentName_Summary_Facet.csv*'.

```
[5]: arena.
      ↪ summarize_facet(cutoffs=(10,70),copy_to_clipboard=True,write_to_csvfile=True);
      ↪
```

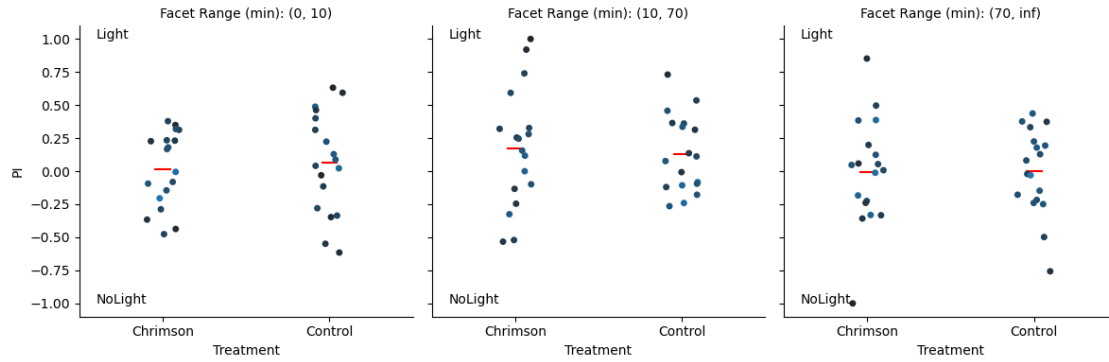
1.6 Plotting

There are lots of cool plotting funtions. Some are shown below.

Facet plots: First of PI, then Percentage, then Total Distance (in mm/min).

Cutoffs separate the acclimaton phase from the experiment phase from the cool down phase.

```
[6]: arena.plot_pi_facet(cutoffs=(10,70))
```



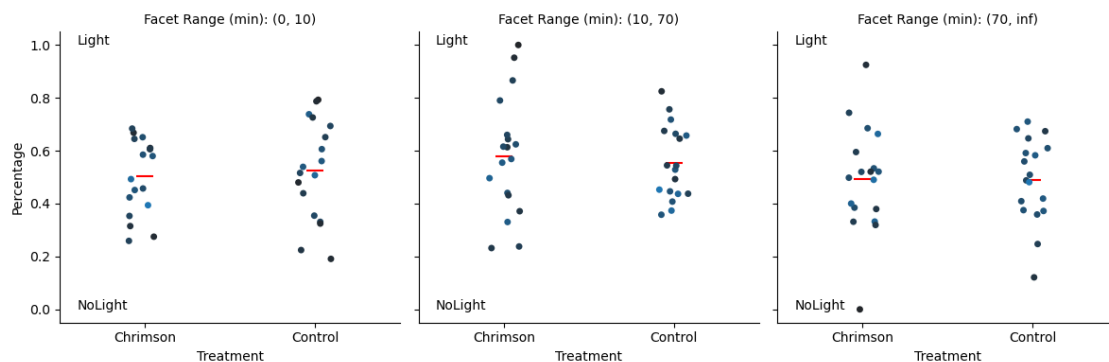
```
[7]: arena.run_pairwise_comparisons_facet(metric="FinalPI",cutoffs=(10,70))
```

```
##### T-Test #####
Column = FinalPI, Range Minutes = (0.00 , 10.00)
Chrimson vs. Control: T=-0.40, p=0.68882
```

```
##### T-Test #####
Column = FinalPI, Range Minutes = (10.00 , 70.00)
Chrimson vs. Control: T=0.34, p=0.73701
```

```
##### T-Test #####
Column = FinalPI, Range Minutes = (70.00 , inf)
Chrimson vs. Control: T=-0.03, p=0.97779
```

```
[8]: arena.plot_percentage_facet(cutoffs=(10,70))
```



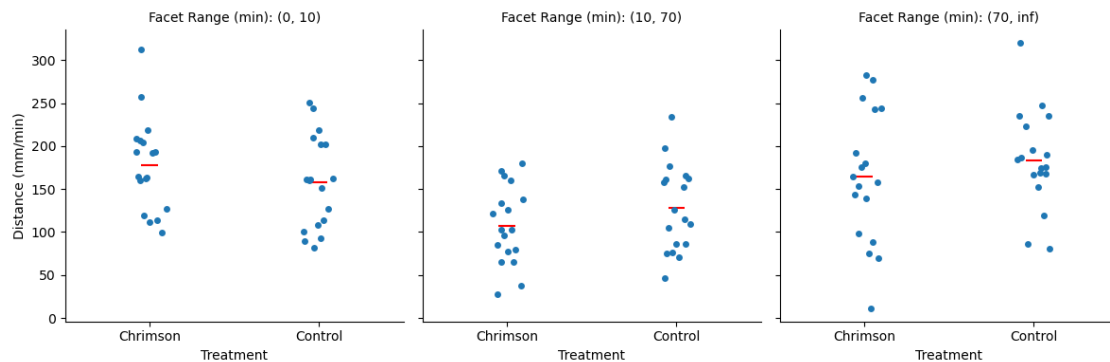
```
[9]: arena.run_pairwise_comparisons_facet(metric="FinalPercentage",cutoffs=(10,70))
```

```
##### T-Test #####
Column = FinalPercentage, Range Minutes = (0.00 , 10.00)
Chrimson vs. Control: T=-0.41, p=0.68646
```

```
##### T-Test #####
Column = FinalPercentage, Range Minutes = (10.00 , 70.00)
Chrimson vs. Control: T=0.42, p=0.68061
```

```
##### T-Test #####
Column = FinalPercentage, Range Minutes = (70.00 , inf)
Chrimson vs. Control: T=0.01, p=0.99575
```

```
[10]: arena.plot_totaldistance_facet(cutoffs=(10,70))
```



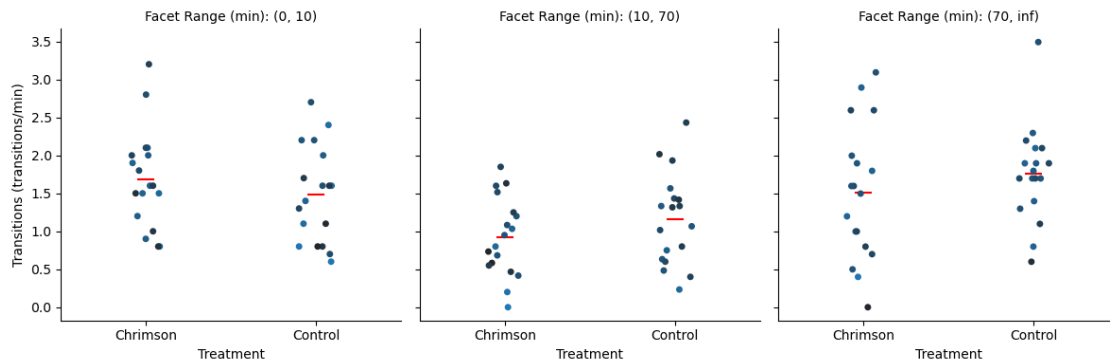
```
[11]: arena.run_pairwise_comparisons_facet(metric="TotalDistance",cutoffs=(10,70))
```

```
##### T-Test #####
Column = TotalDistance, Range Minutes = (0.00 , 10.00)
Chrimson vs. Control: T=1.15, p=0.25919
```

```
##### T-Test #####
Column = TotalDistance, Range Minutes = (10.00 , 70.00)
Chrimson vs. Control: T=-1.27, p=0.21341
```

```
##### T-Test #####
Column = TotalDistance, Range Minutes = (70.00 , inf)
Chrimson vs. Control: T=-0.87, p=0.39035
```

```
[12]: arena.plot_transitions_facet(cutoffs=(10,70))
```



```
[13]: arena.run_pairwise_comparisons_facet(metric="Transitions",cutoffs=(10,70))
```

```
##### T-Test #####
```

```
Column = Transitions, Range Minutes = (0.00 , 10.00)
```

```
Chrimson vs. Control: T=0.96, p=0.34293
```

```
##### T-Test #####
```

```
Column = Transitions, Range Minutes = (10.00 , 70.00)
```

```
Chrimson vs. Control: T=-1.25, p=0.21999
```

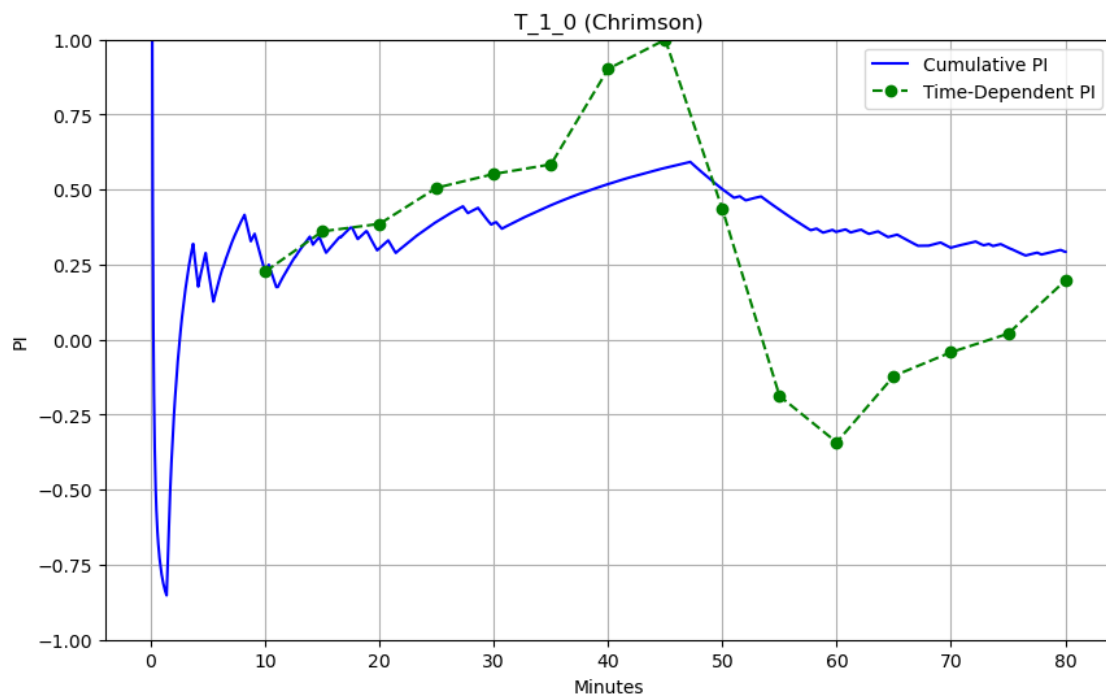
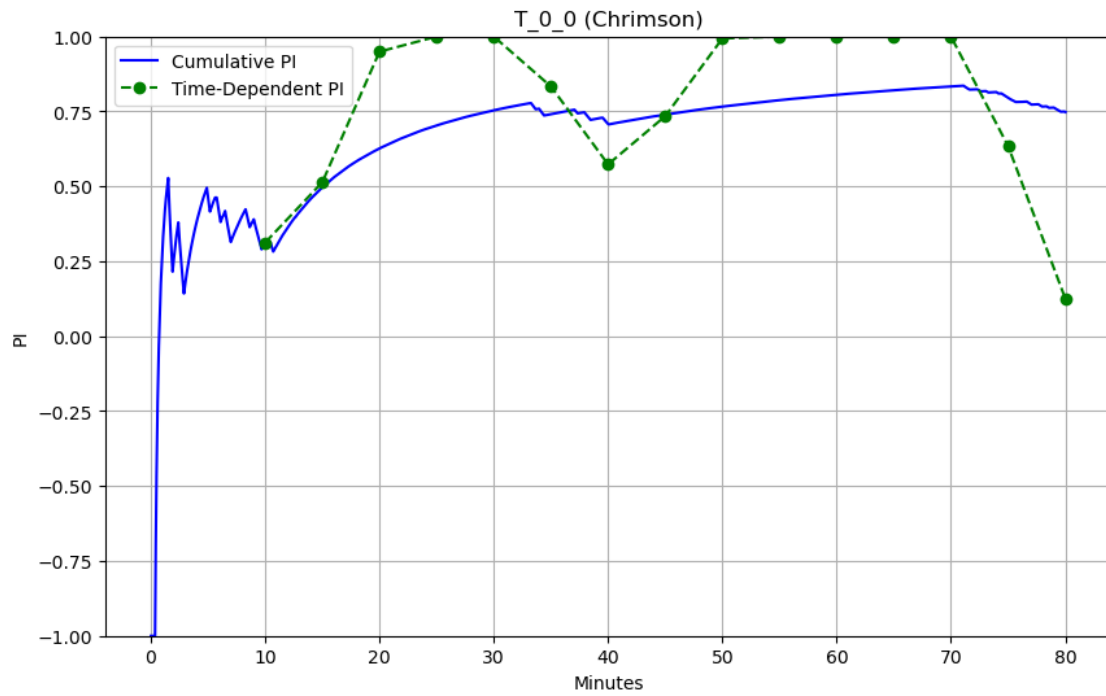
```
##### T-Test #####
```

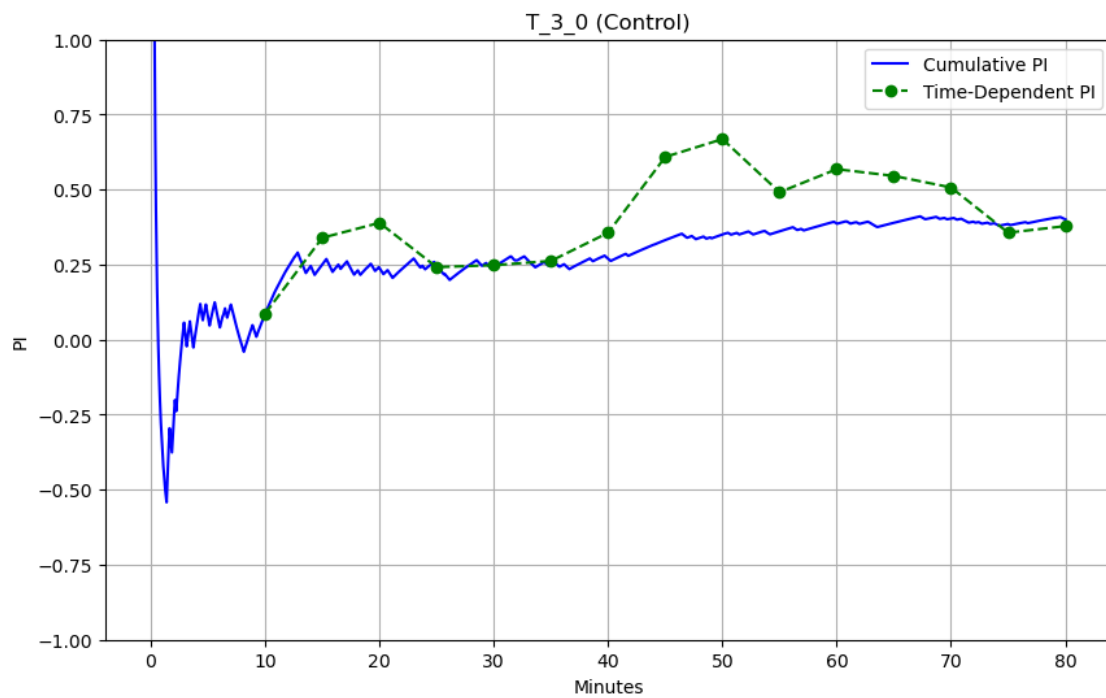
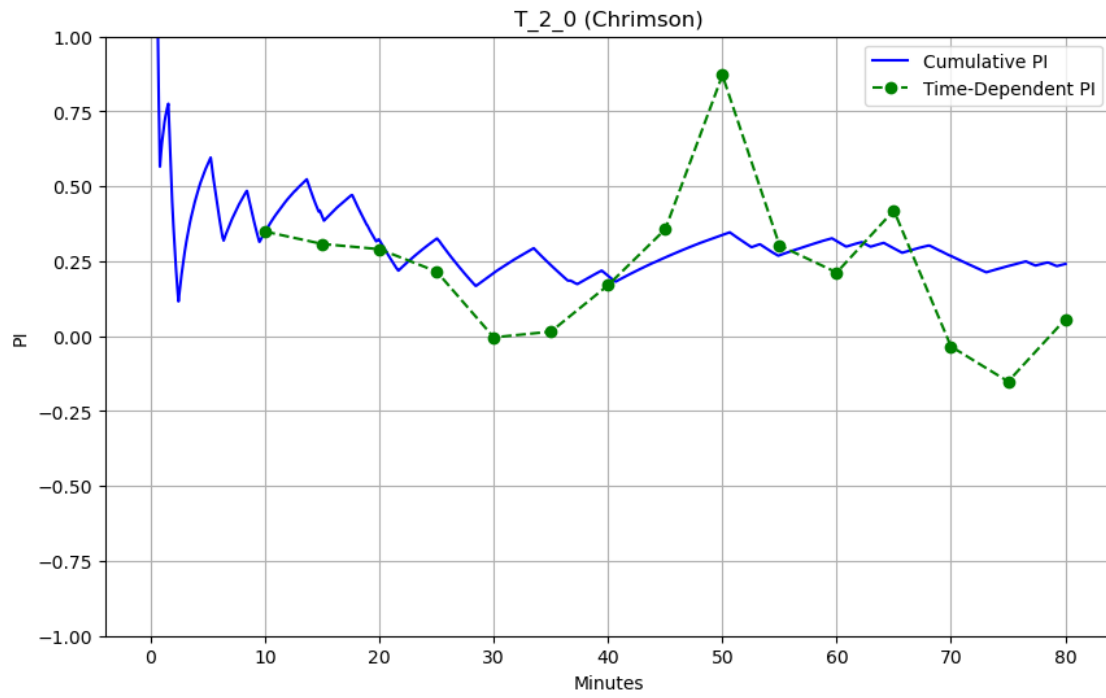
```
Column = Transitions, Range Minutes = (70.00 , inf)
```

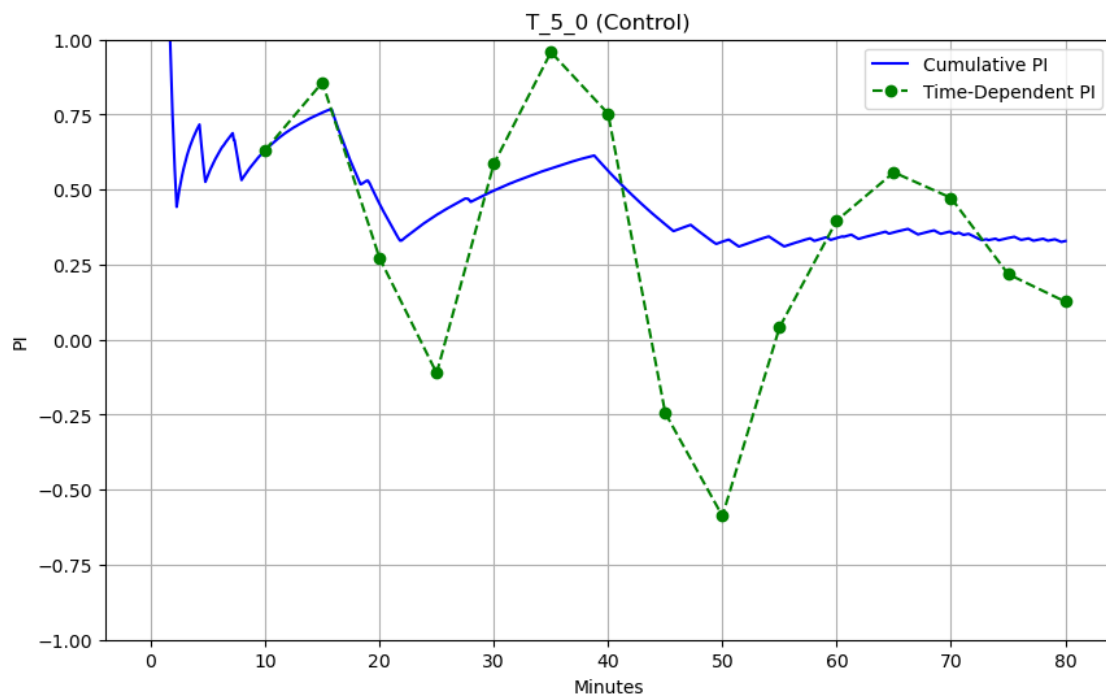
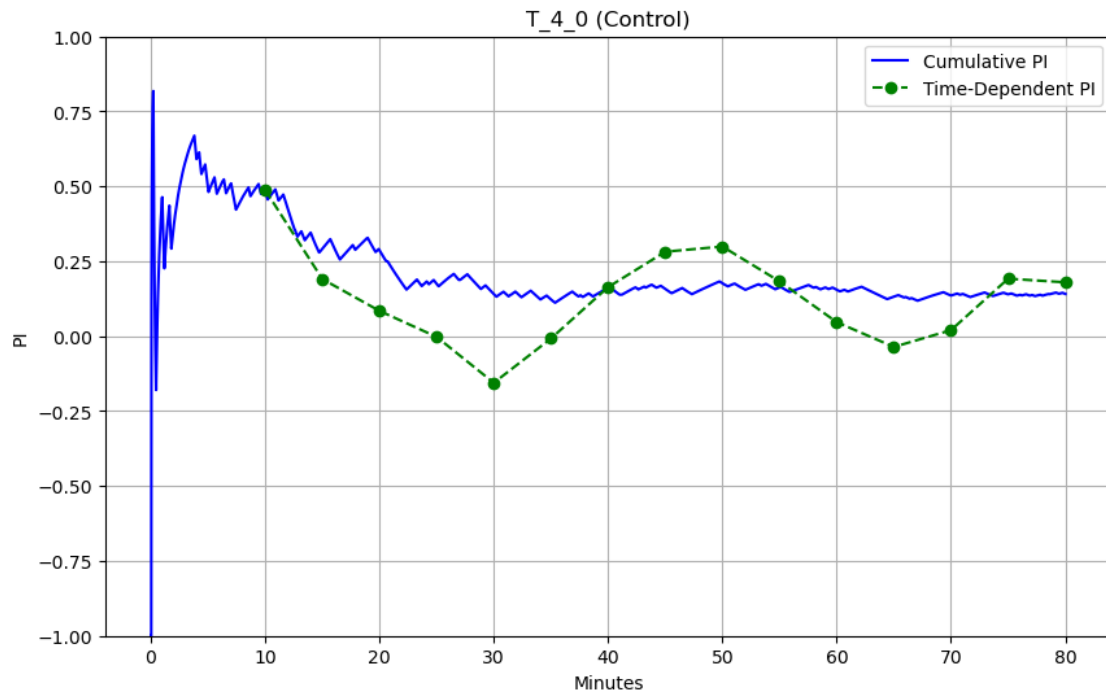
```
Chrimson vs. Control: T=-0.97, p=0.34093
```

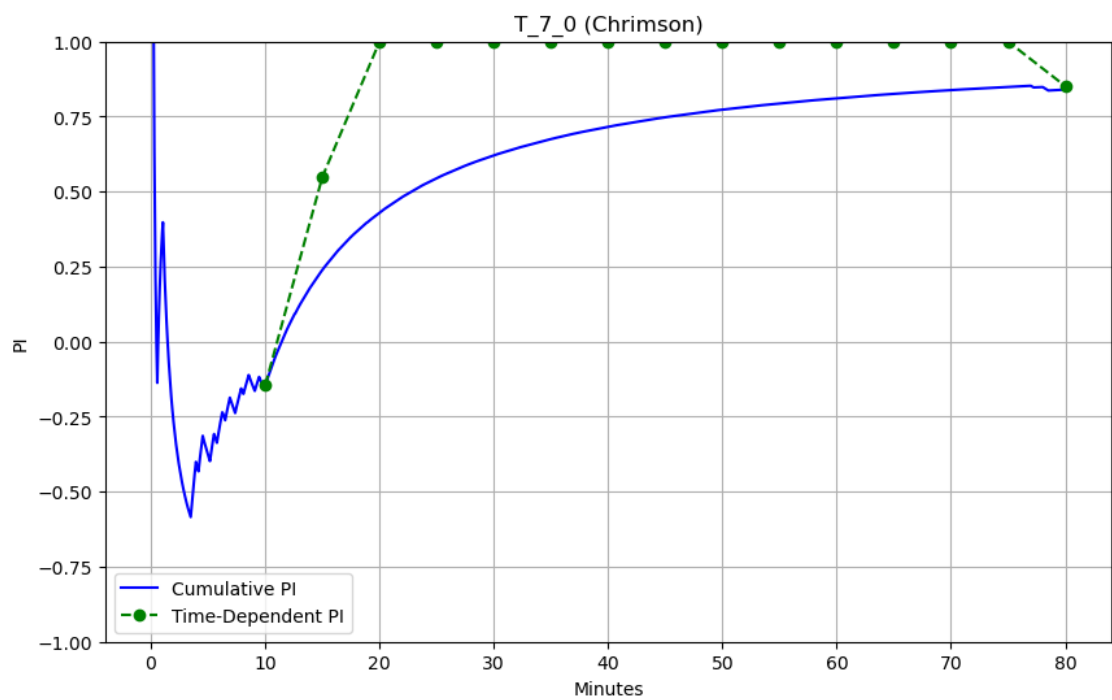
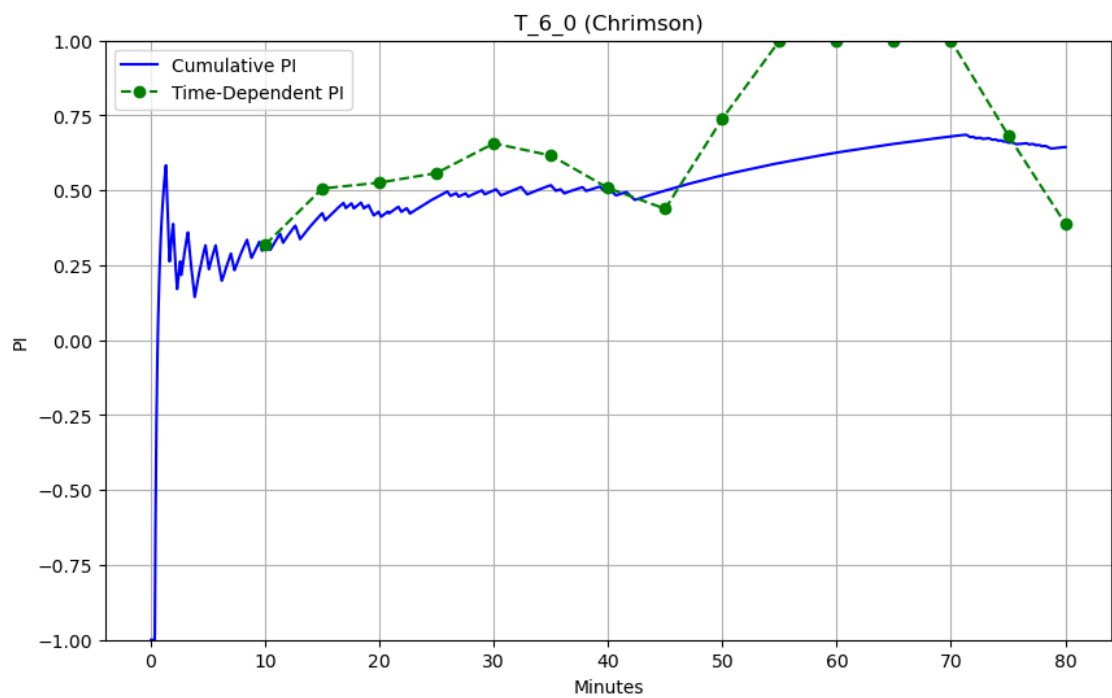
Tracker plots: These show, for each tracker individually, the progression of PI and Percentage showing cumulative values and time-dependent values

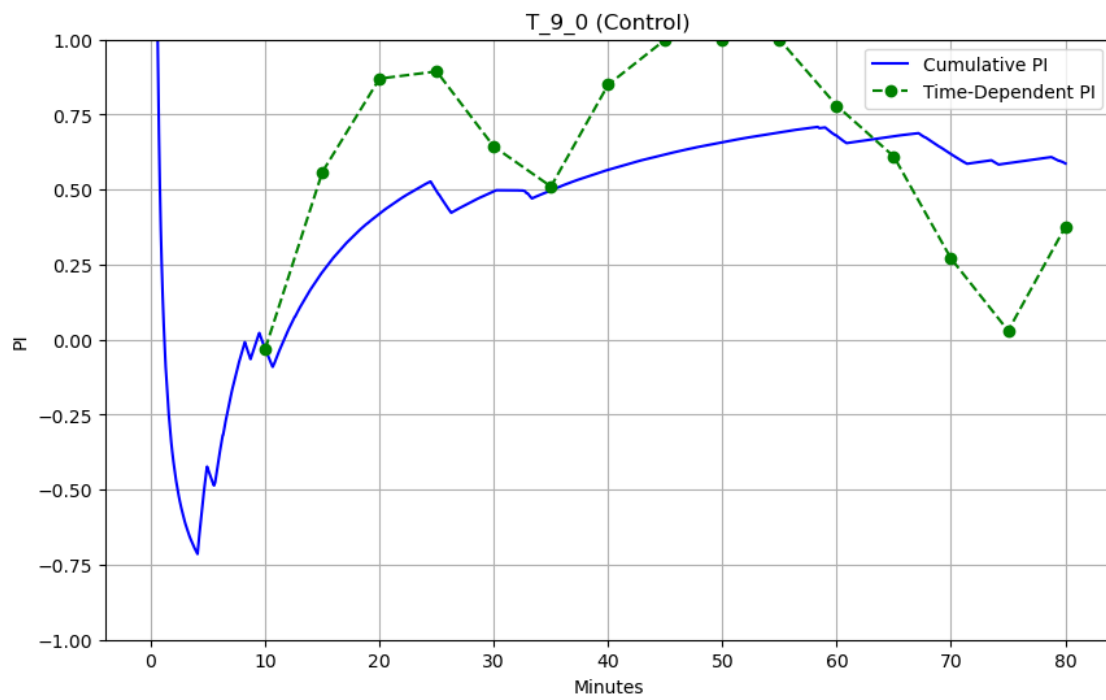
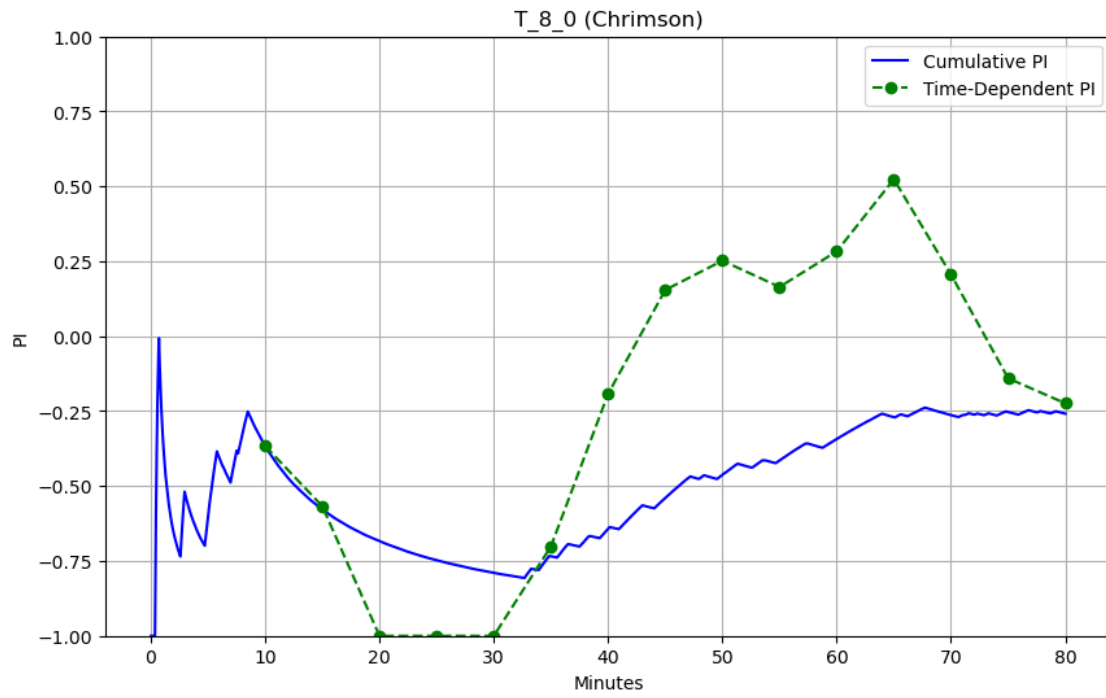
```
[14]: arena.plot_trackers_pis()
```

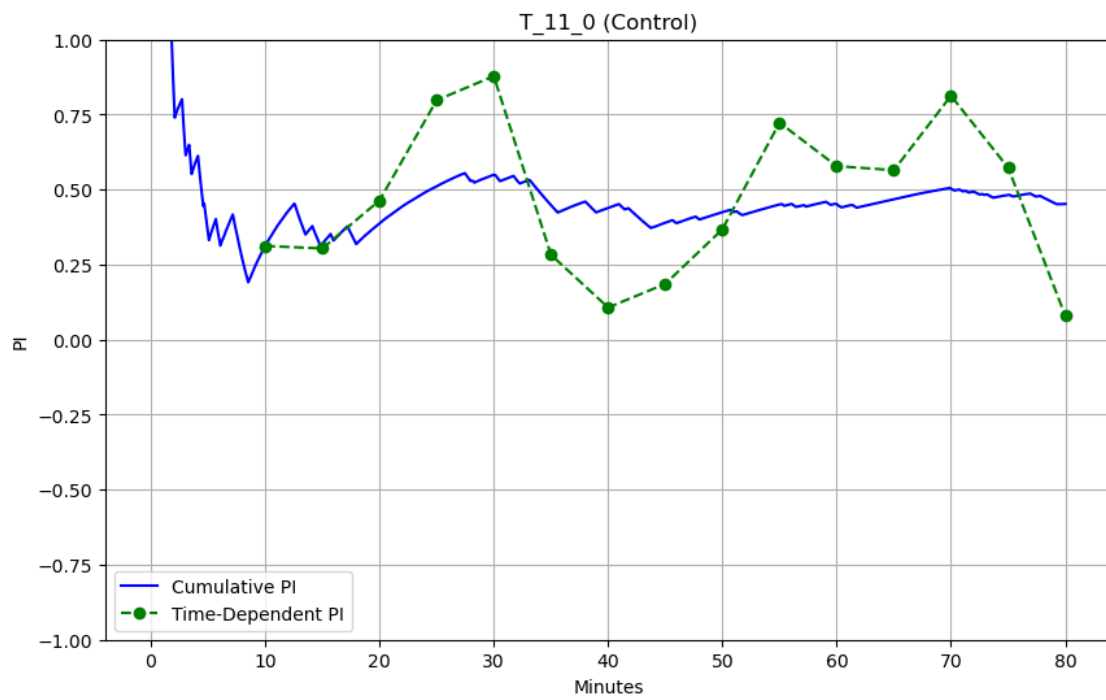
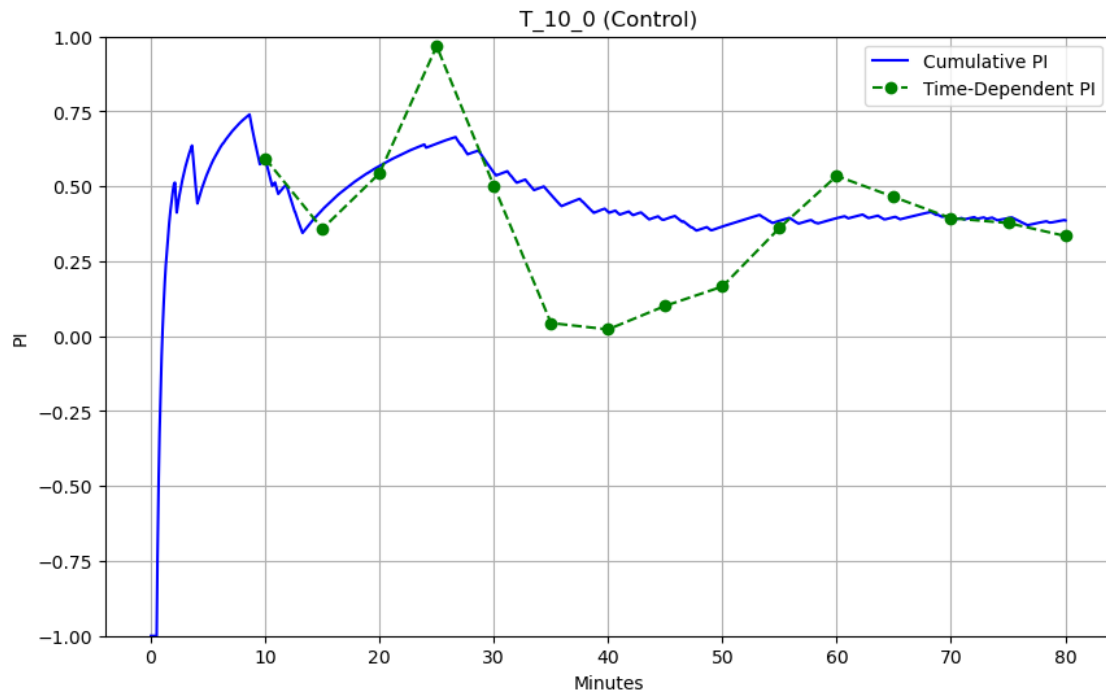


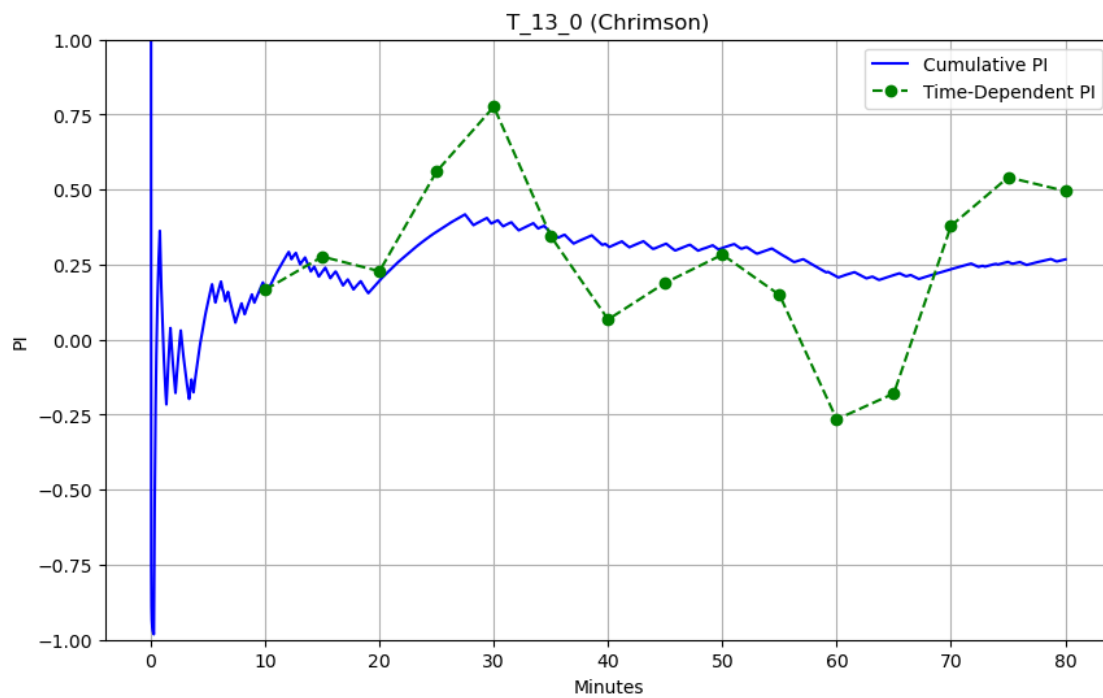
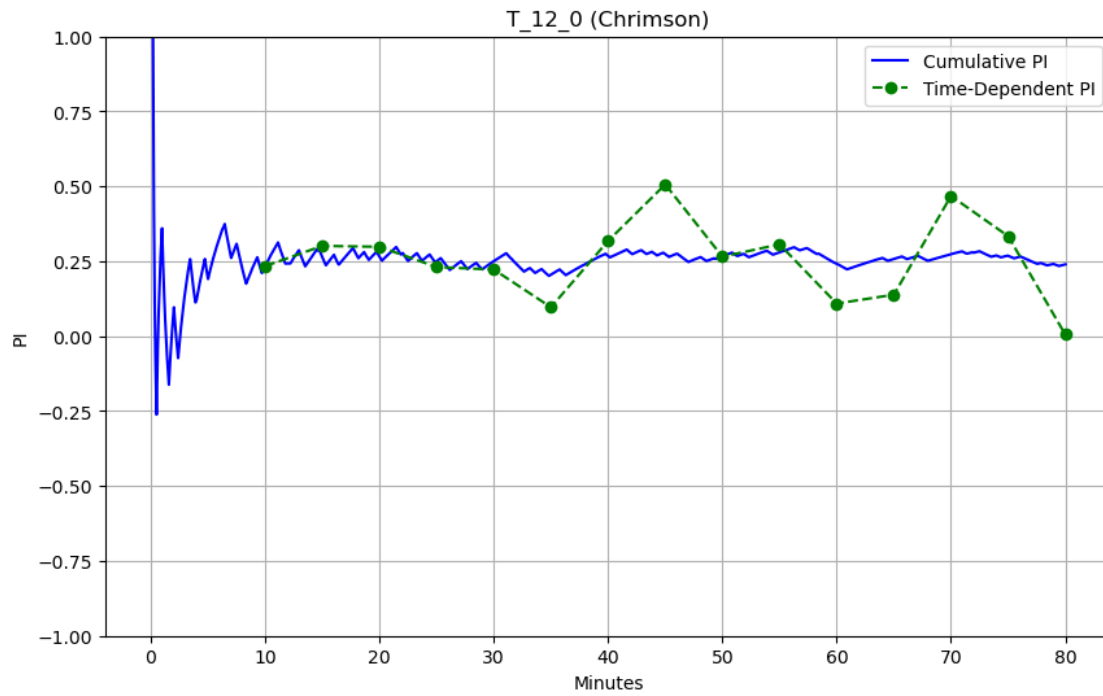


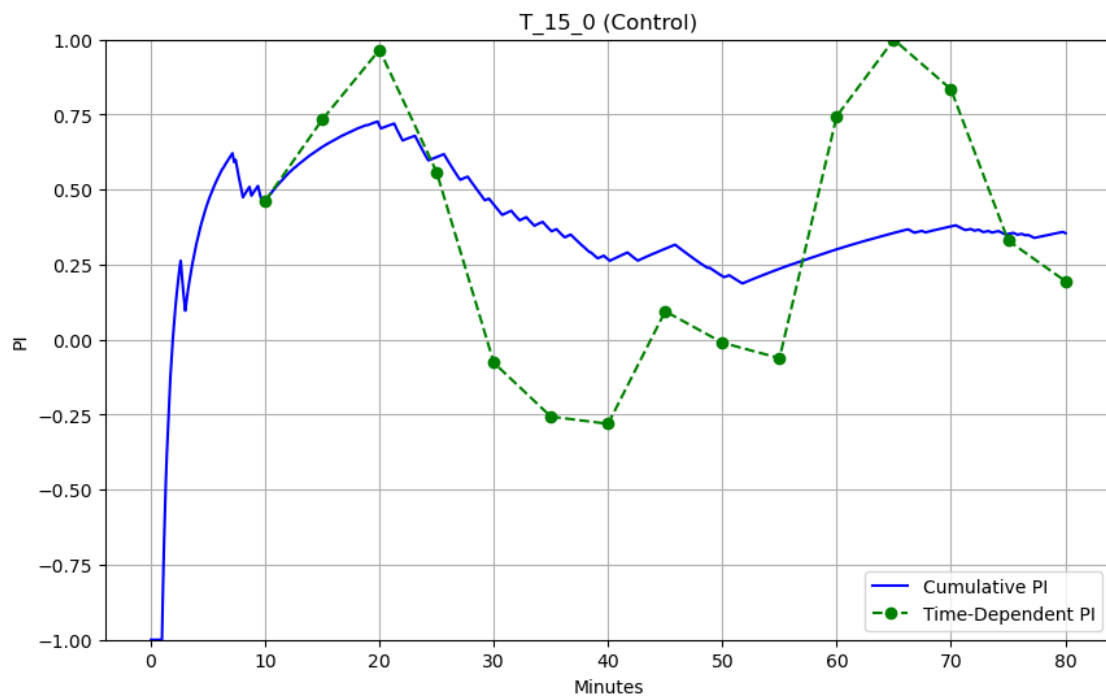
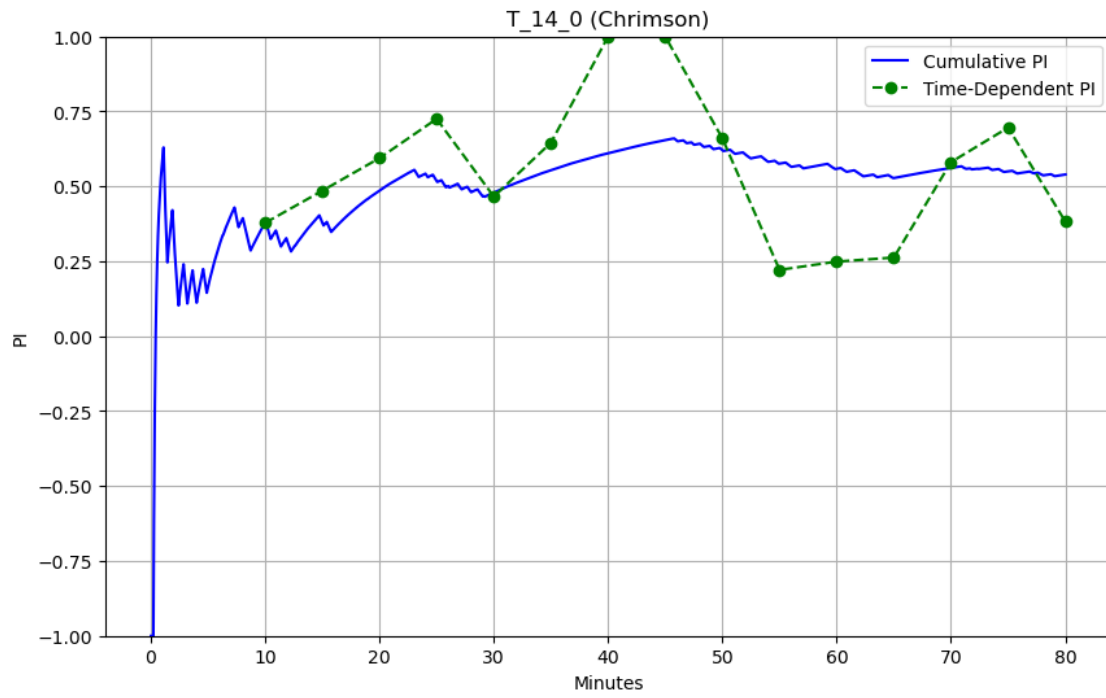


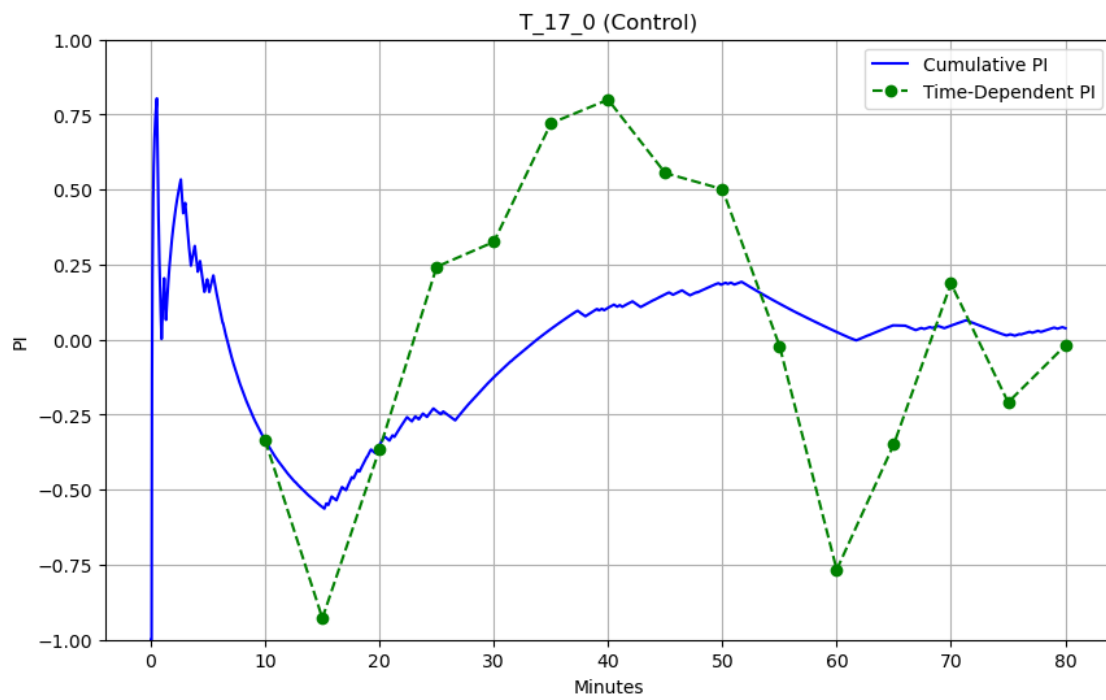
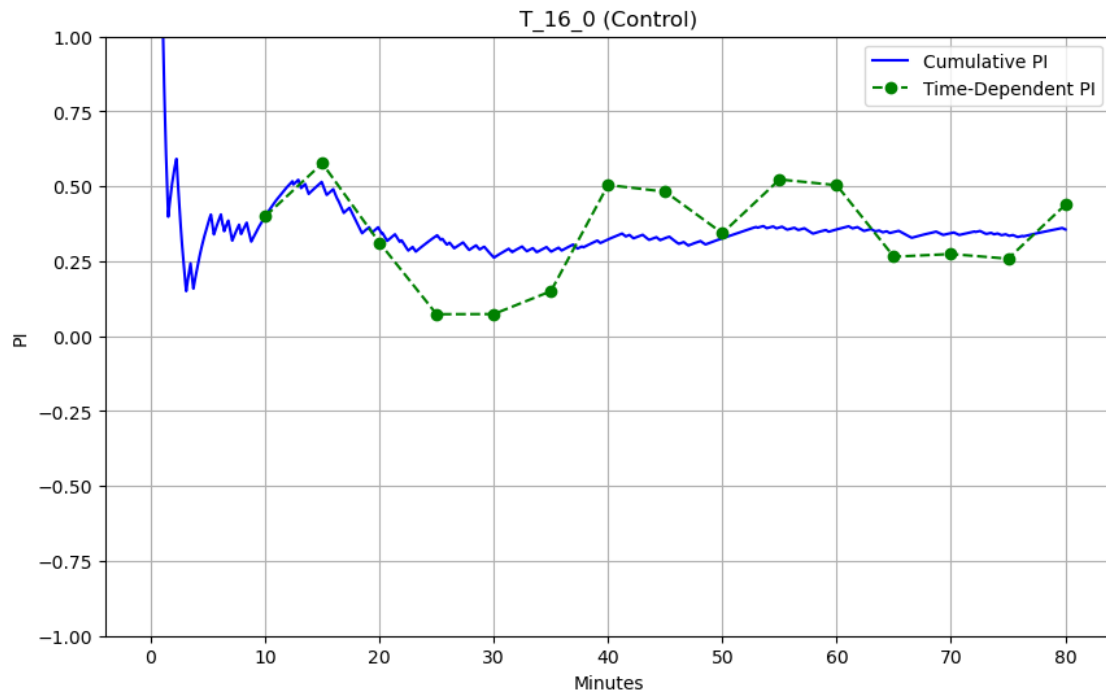


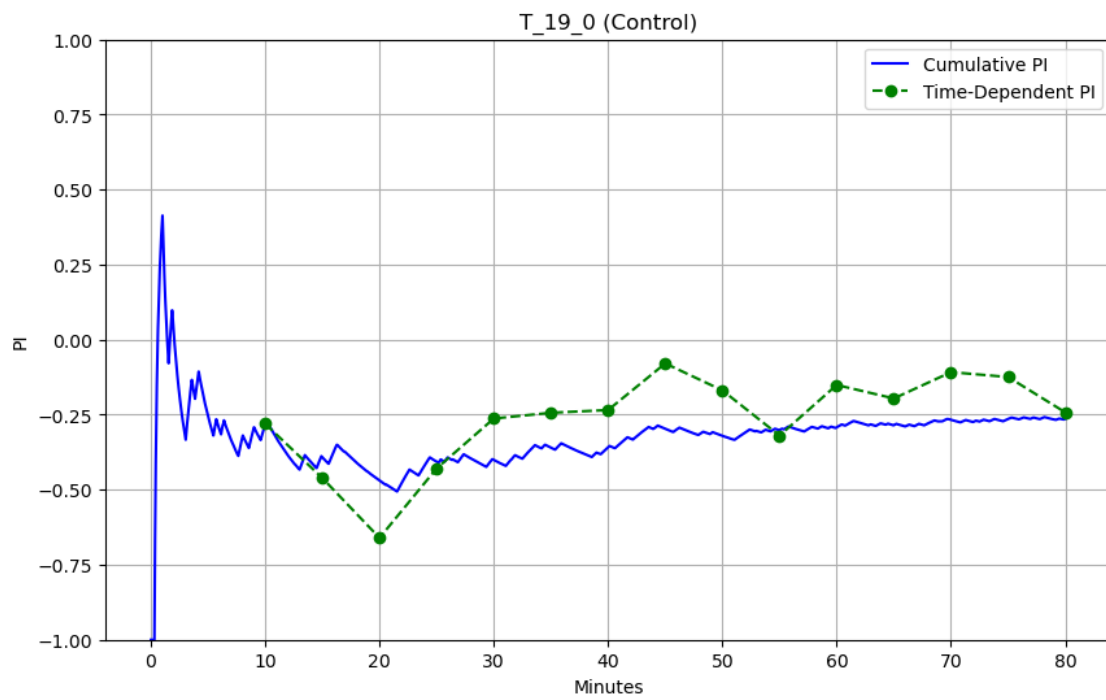
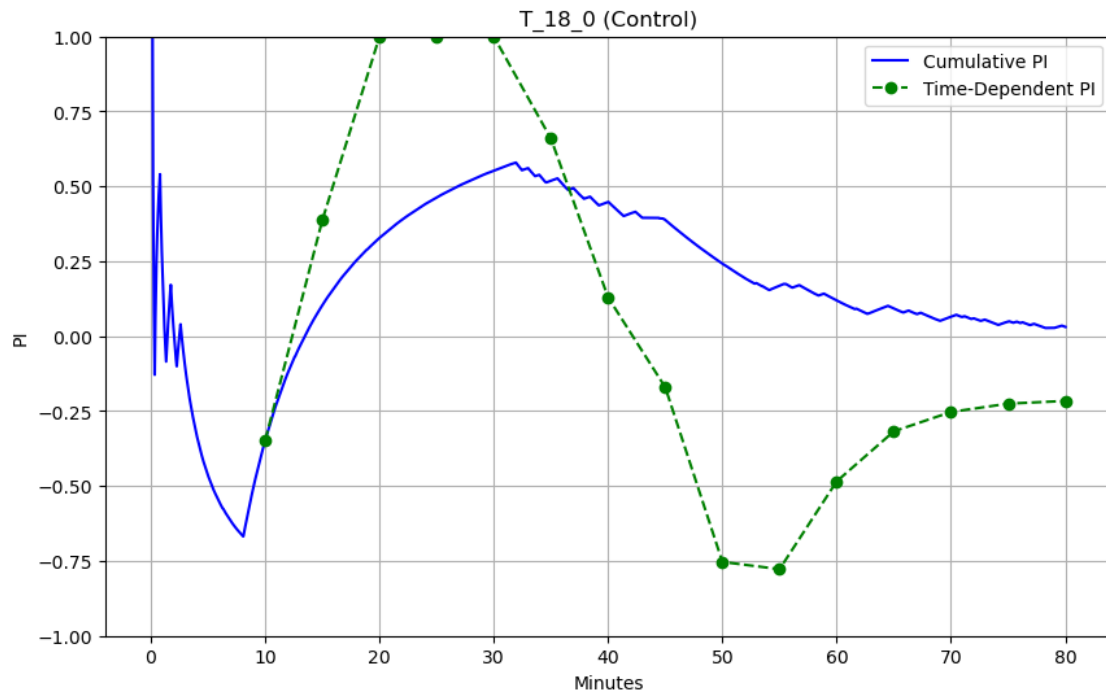


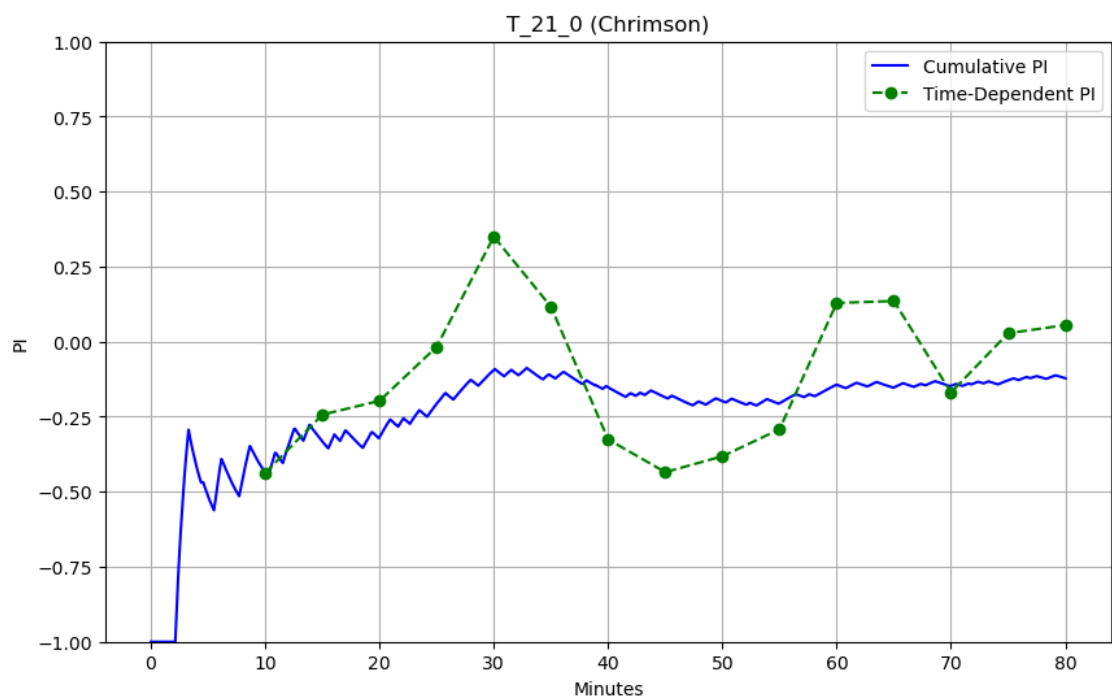
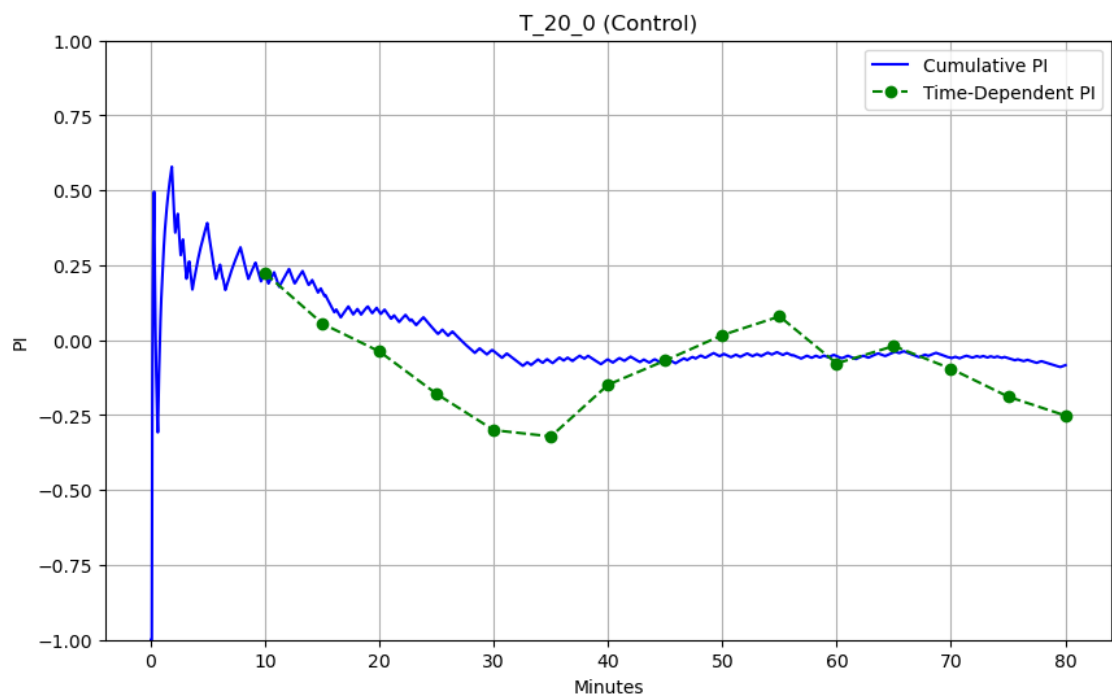


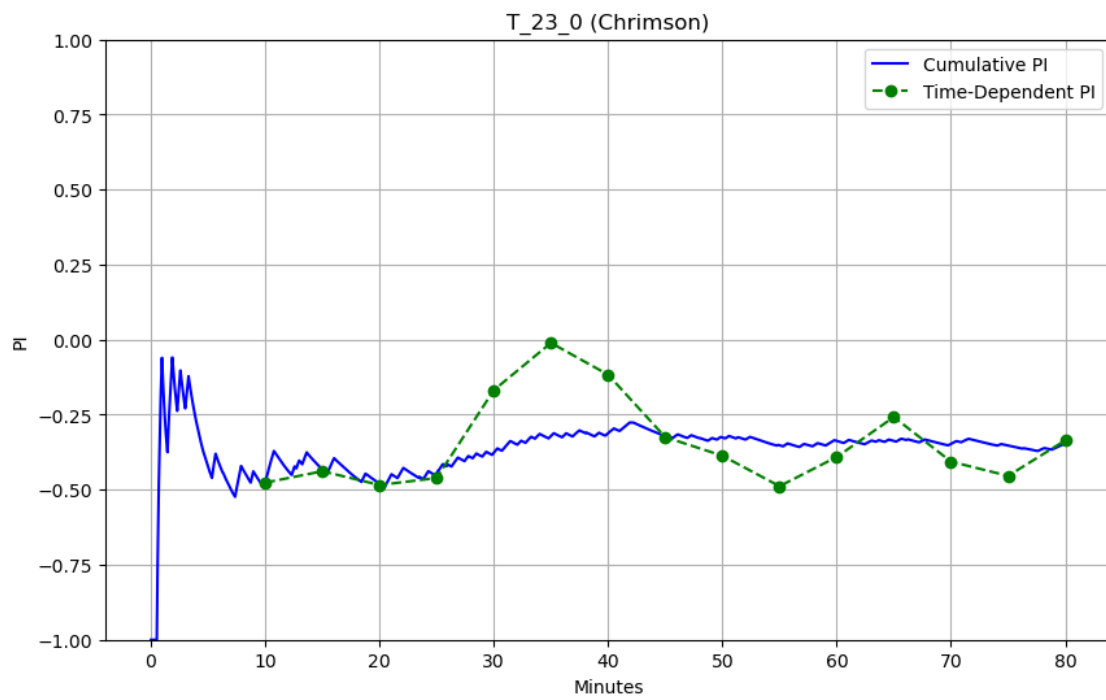
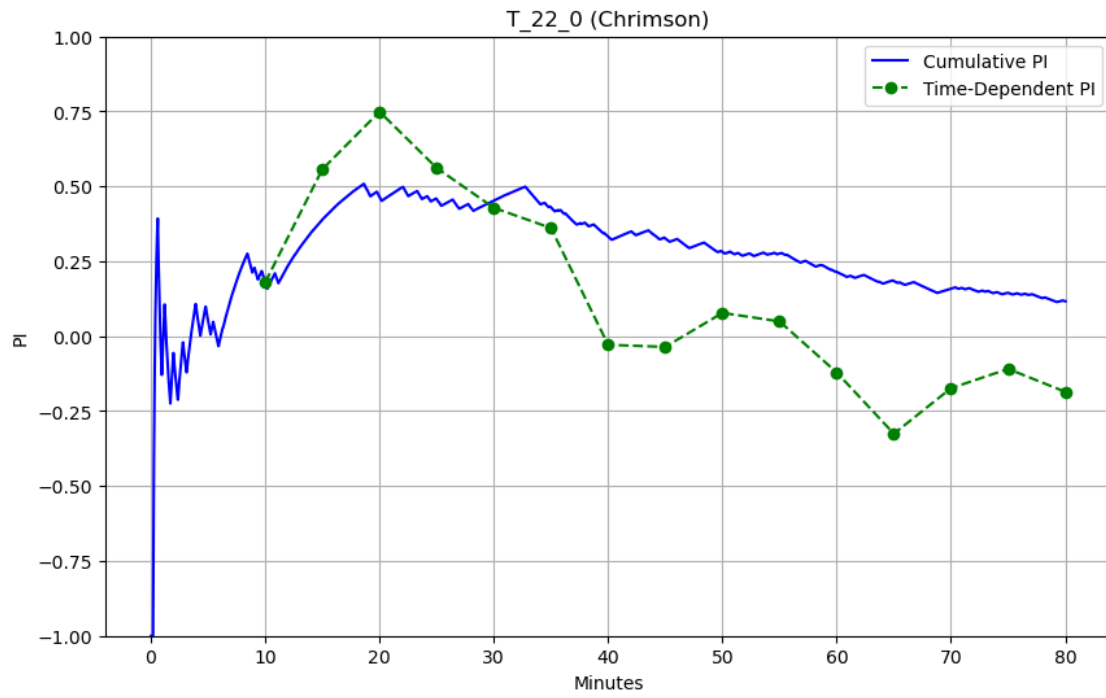


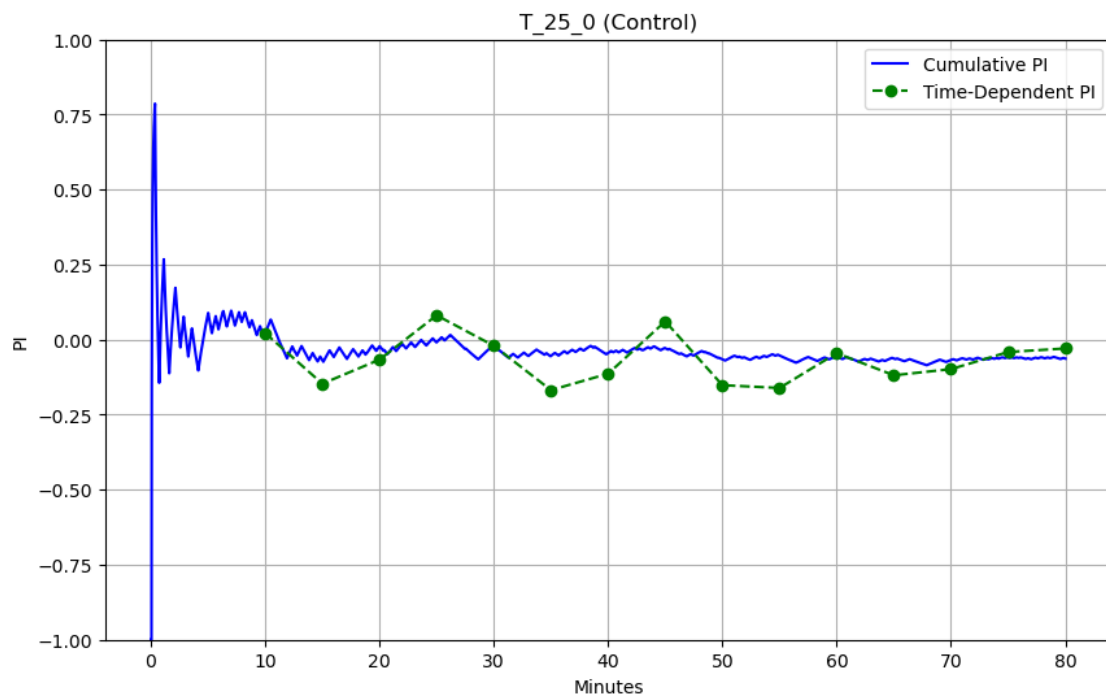
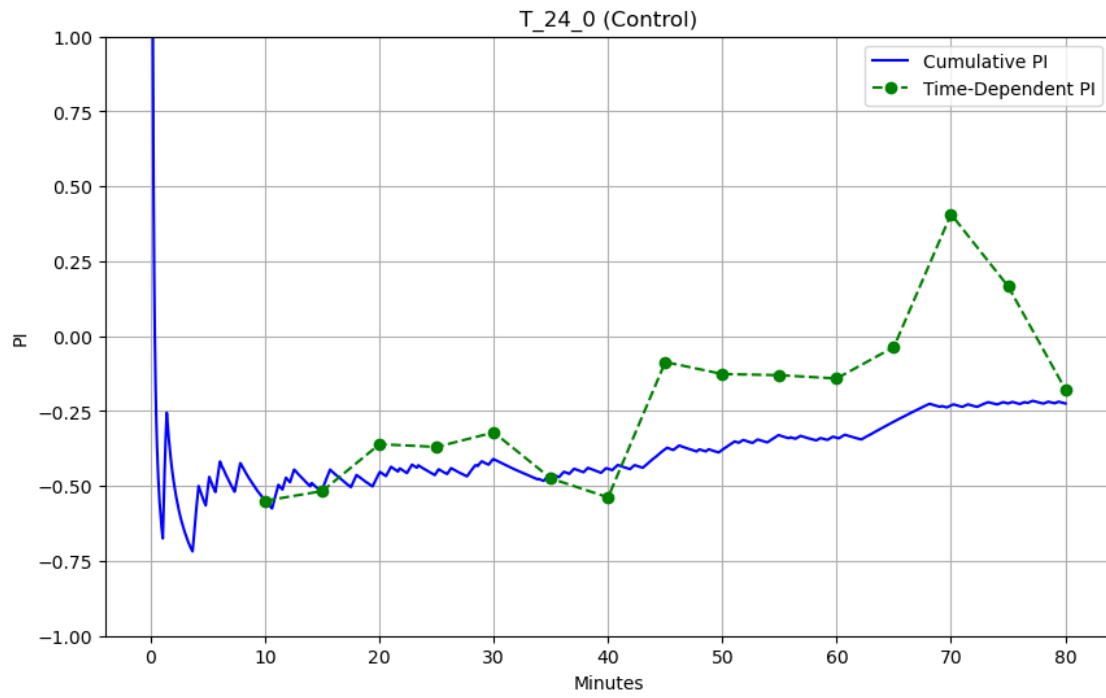


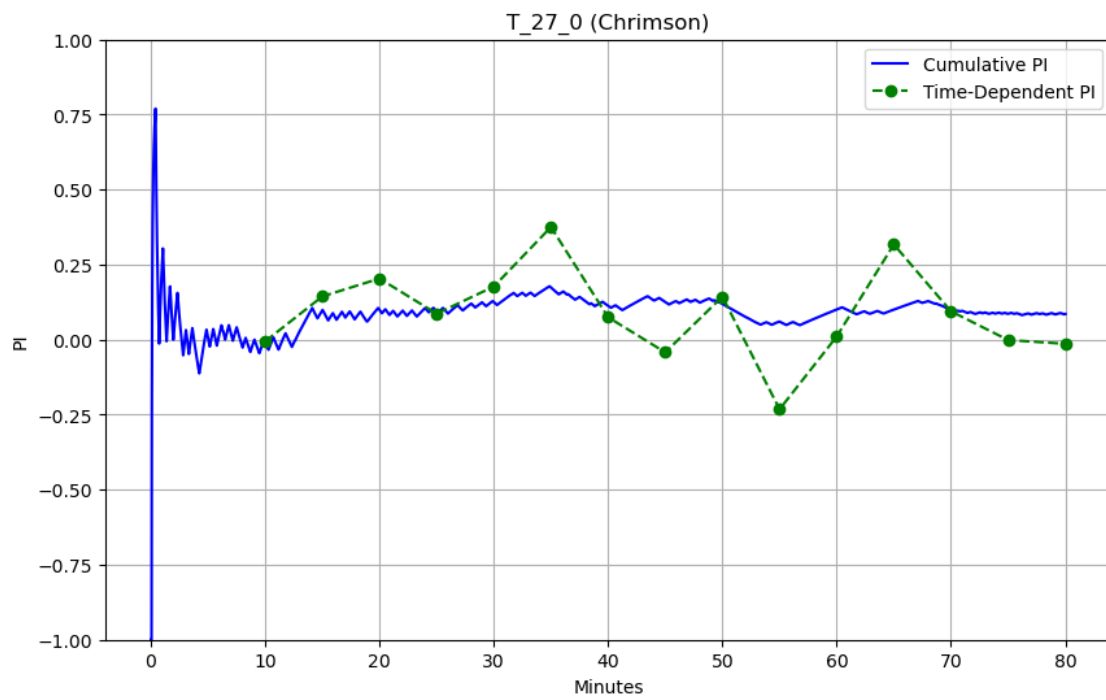
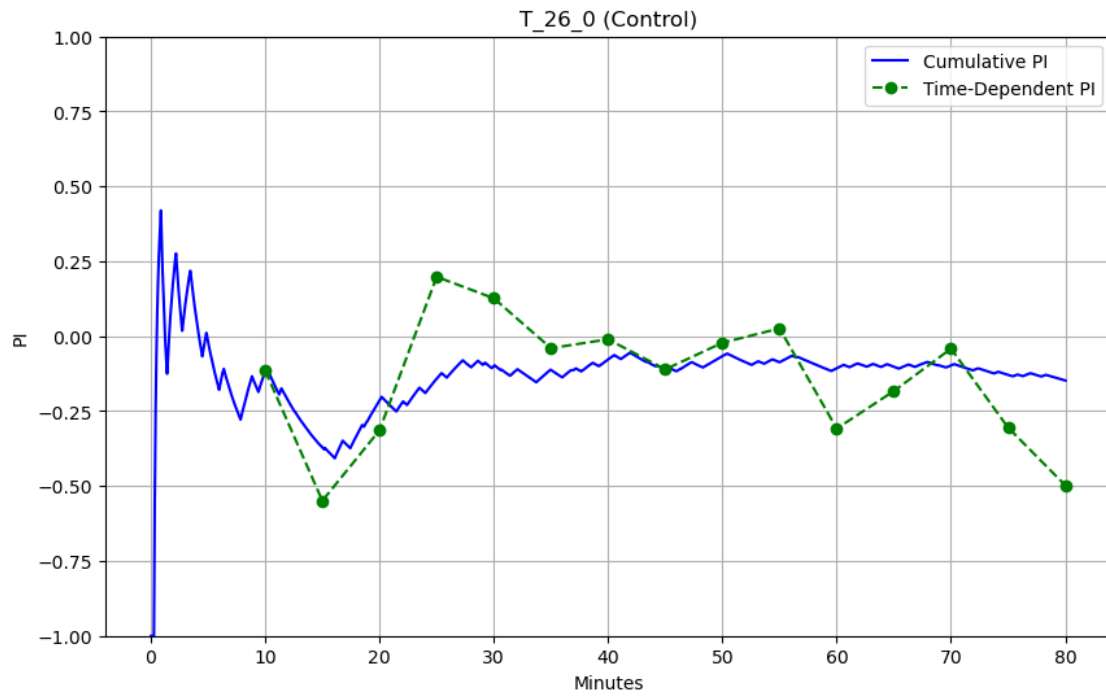


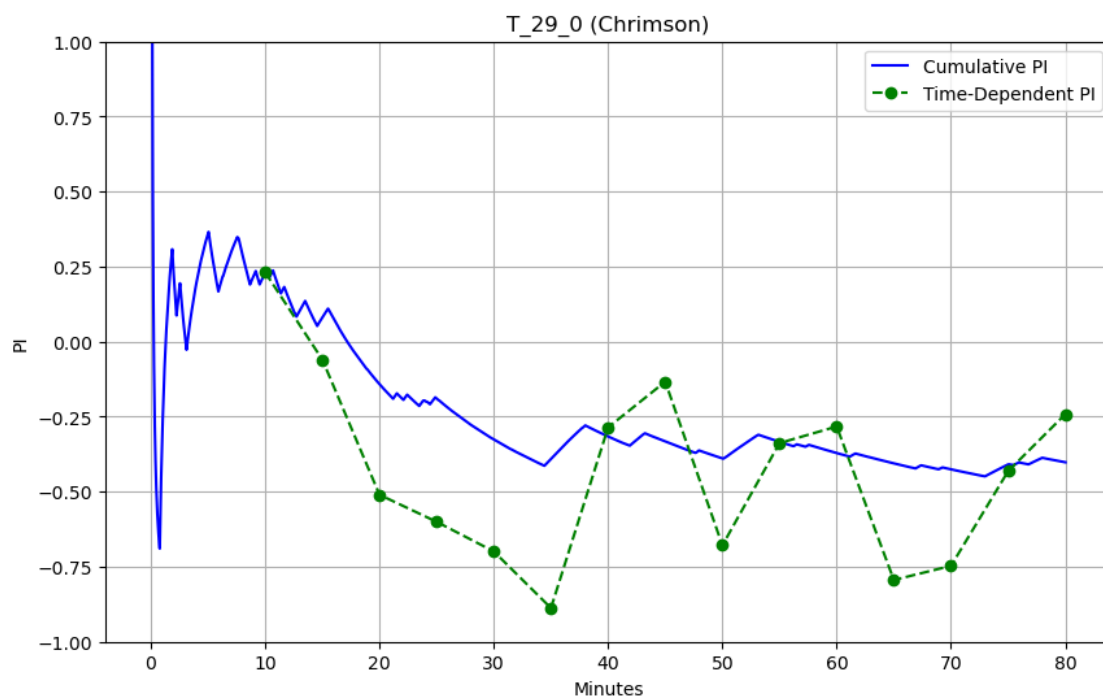
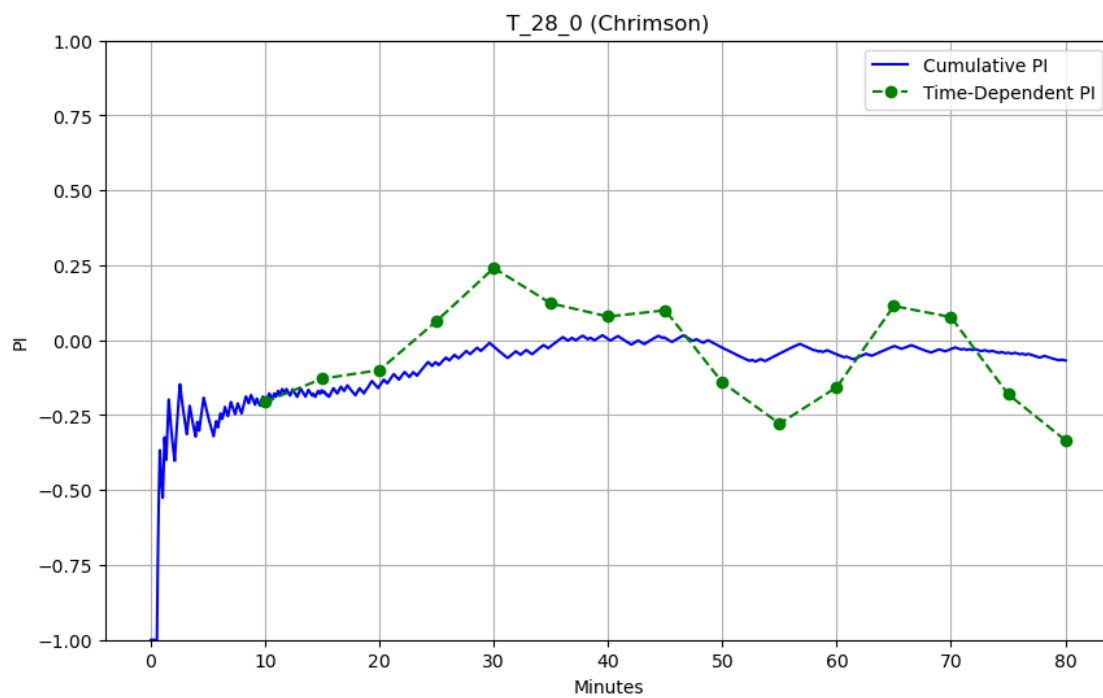


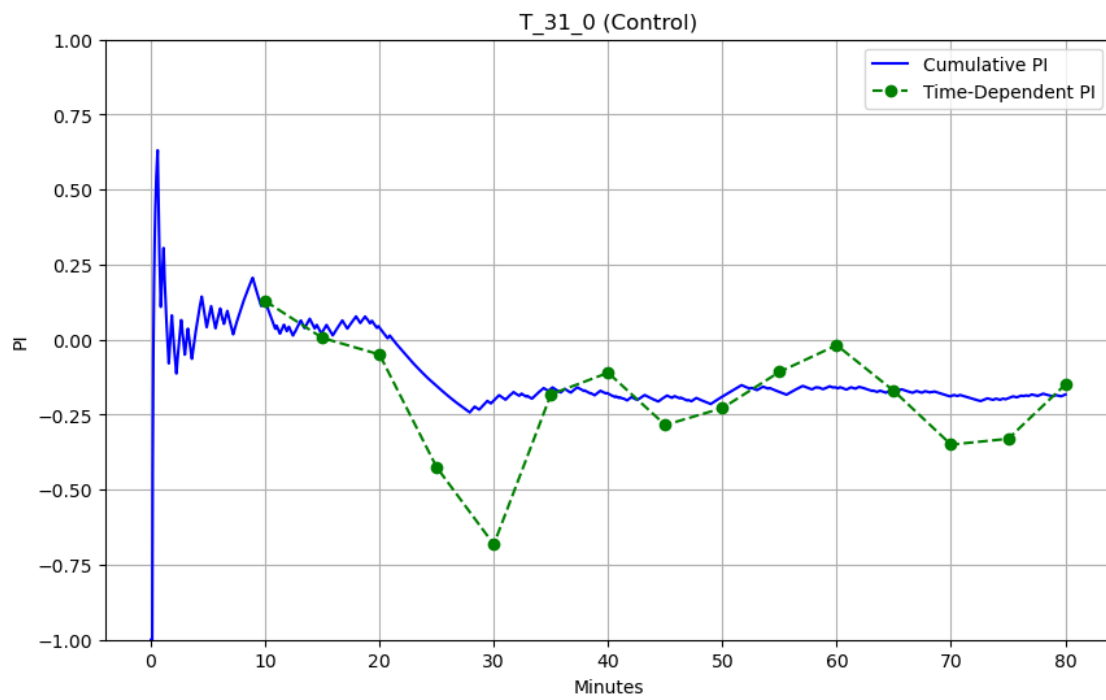
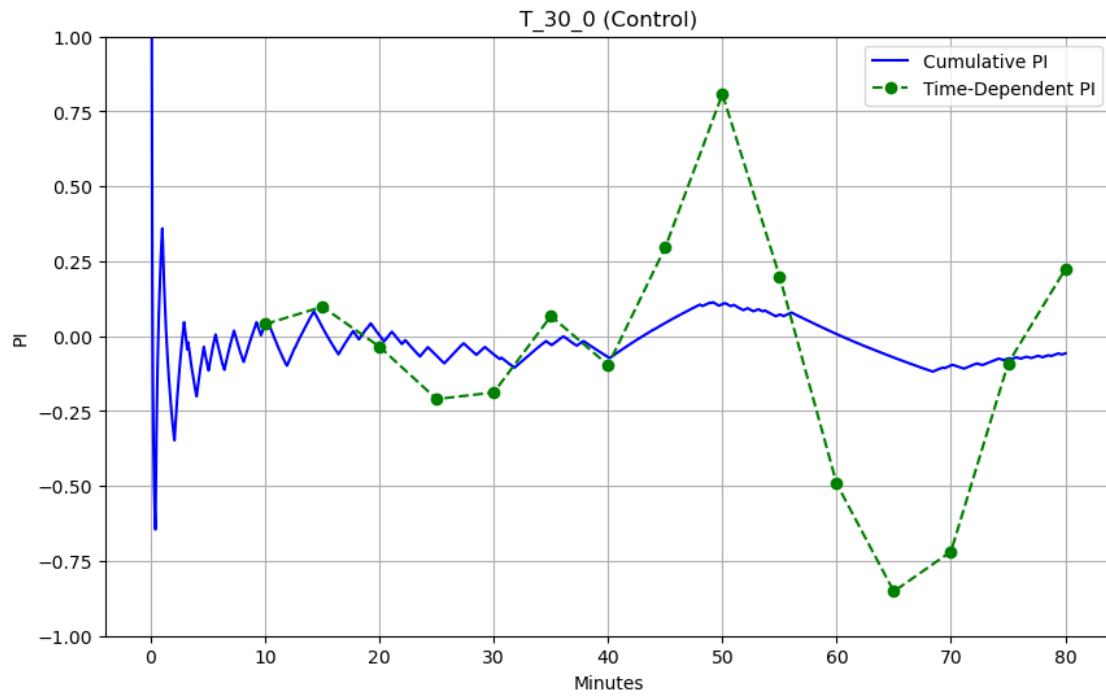


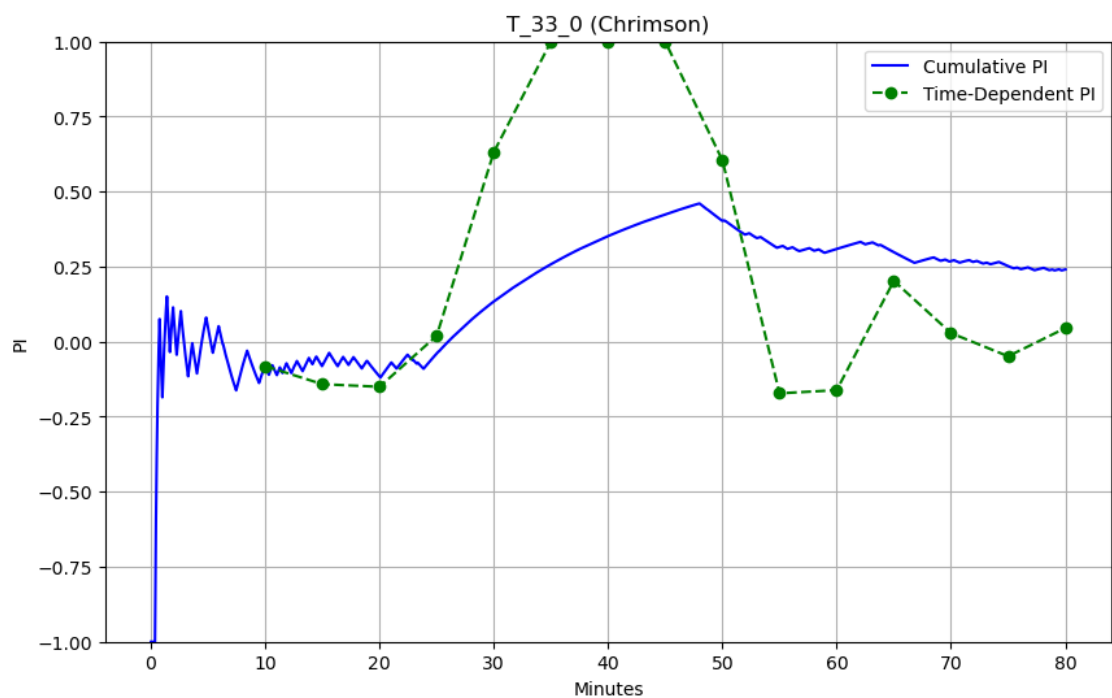
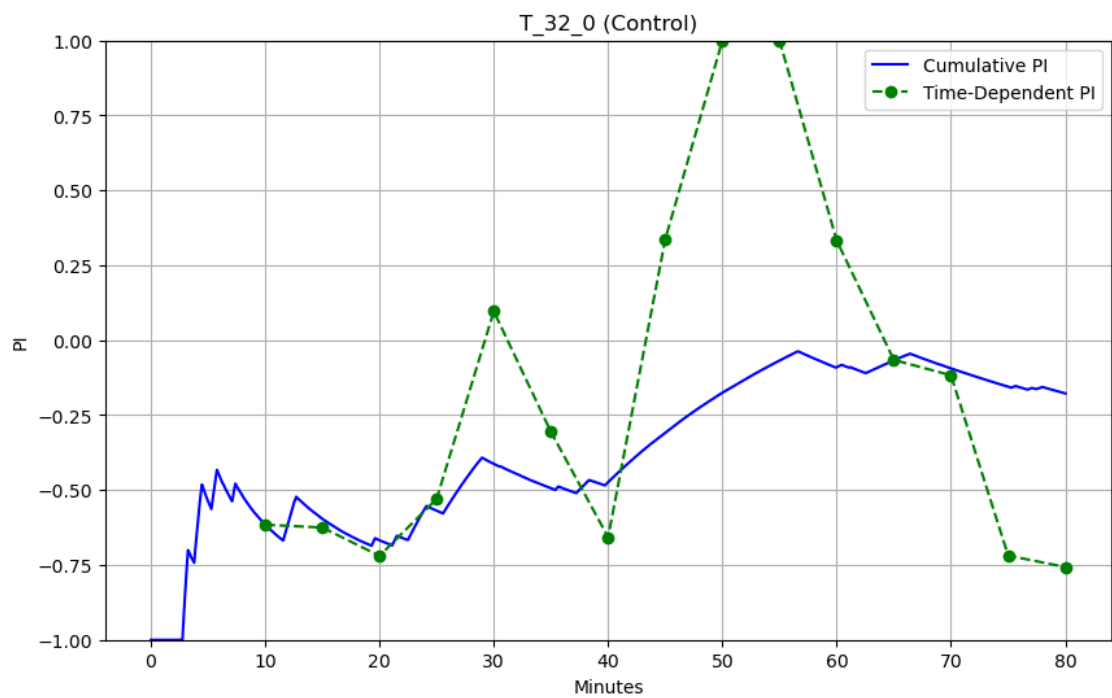


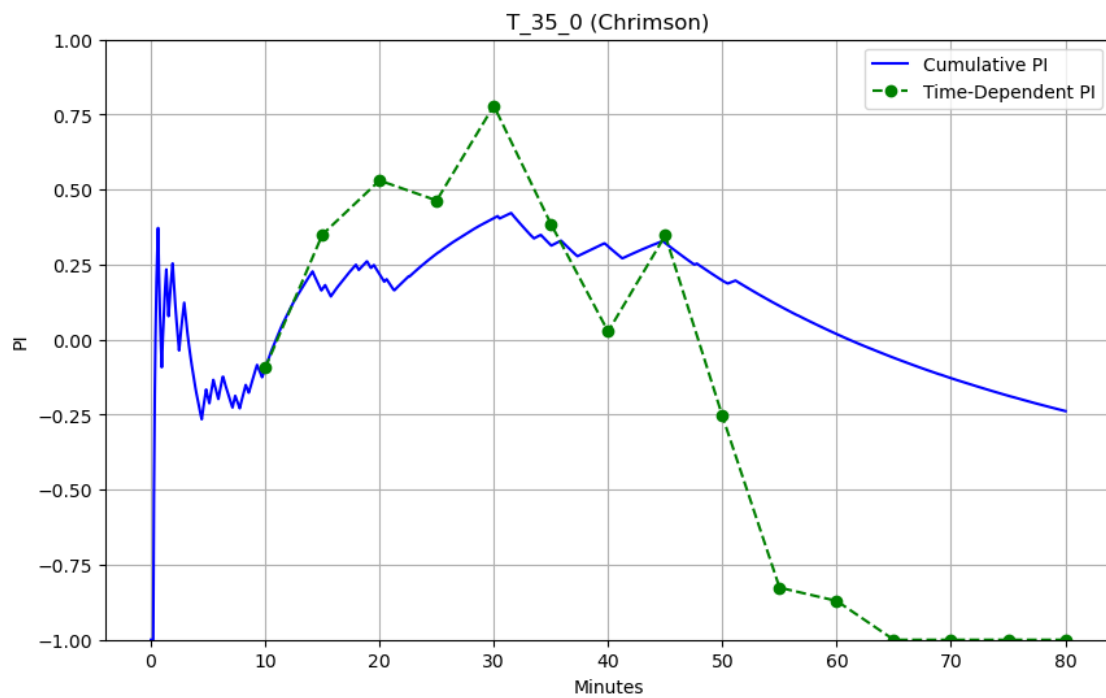
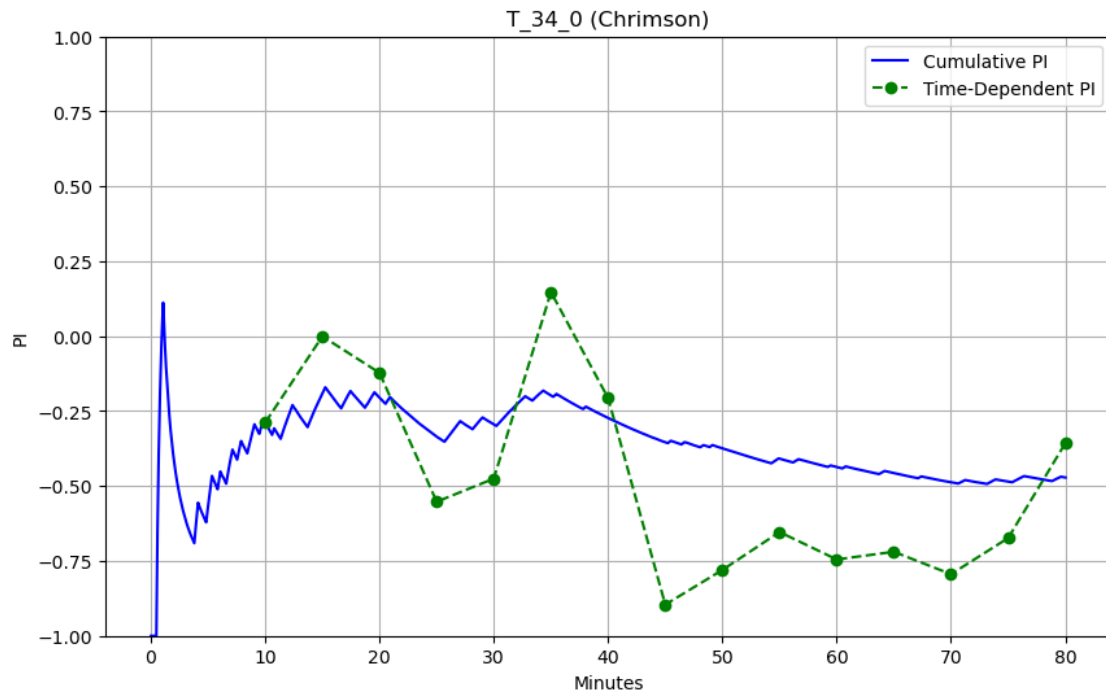






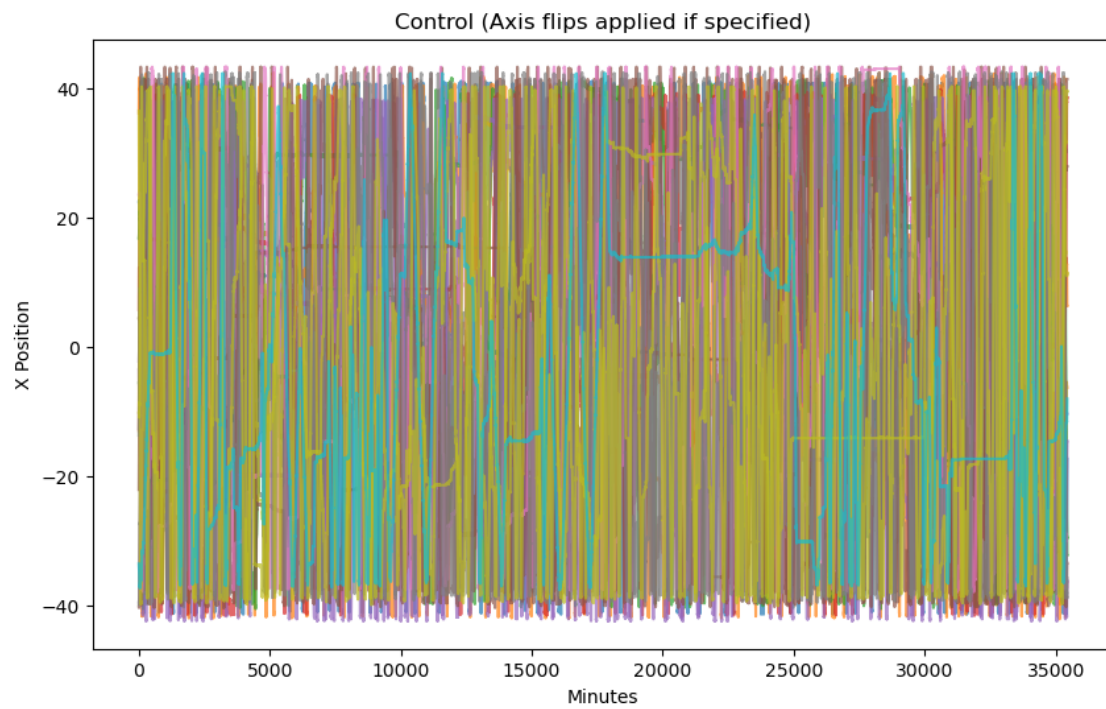
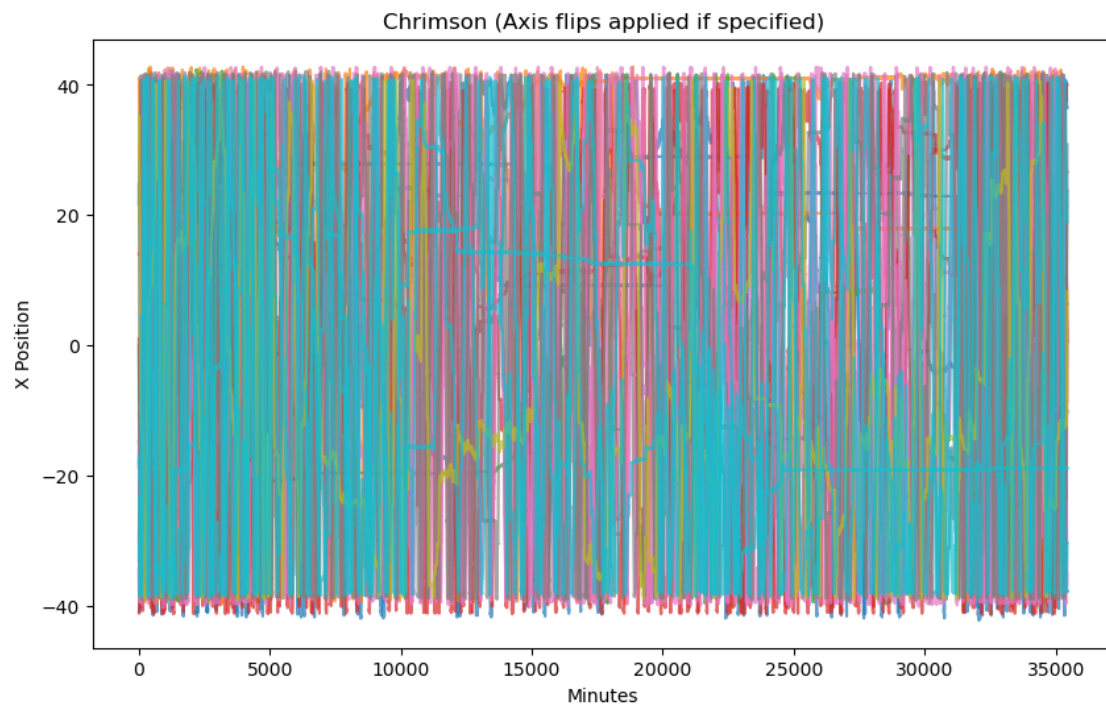






Positional Plots: Show the X positions of each tracker as a function of time.


```
[15]: arena.plot_trackers_x(one_plot=True)
```



```
[16]: arena.plot_trackers_x()
```

